

The Moving Target

External Force:  
THI > 68

Stress:  
Noise/Stray  
Voltage

Panting/Vasodilation

Geometric Shift

Kicking/Shifting

Altered Teat  
Distance

Unstable Target  
Coordinates  
X, Y, Z Shifting

Activates

Teat Geometry  
Bottle, Funnel,  
Flat, Concave

HPA Axis  
Activation

Secretion

Cortisol Release

Vasoconstriction

Oxytocin  
Antagonism  
Let-down  
Inhibition

Let-down Inhibited

Normal

FME:  
Incomplete  
Milking

Event

Robotic Arm Dynamics

Arm Dynamics  
Flexibility vs.  
Weight

Detection  
System  
Target  
Acquisition

Liner Technology  
e.g., Stimulor  
StressLess

Target Alignment

Aligned

Arm too slow / Cow too fast

FME: Attachment  
Failure

Liner Interaction

Vacuum Leak / Pinching

FME: Cluster  
Detachment

Good Seal

Milk Extraction